

KENDRIYA VIDYALAYA SANGATHAN CHENNAI REGION

PRE-BOARD EXAMINATION 2024-25

CLASS X

SUBJECT: ARTIFICIAL INTELLIGENCE (417)

Max Time : 2 hrs

Max Mark: 50

General Instructions:

1. Please read the instructions carefully.
2. This Question Paper consists of 21 questions in two sections: Section A & Section B.
3. Section A has Objective type questions whereas Section B contains Subjective type questions.
4. Out of the given (5 + 16 =) 21 questions, a candidate has to answer (5 + 10 =) 15 questions in the allotted (maximum) time of 2 hours.
5. All questions of a particular section must be attempted in the correct order.

6. SECTION A - OBJECTIVE TYPE QUESTIONS (24 MARKS):

- i. This section has 05 questions.
- ii. Marks allotted are mentioned against each question/part.
- iii. There is no negative marking.
- iv. Do as per the instructions given.

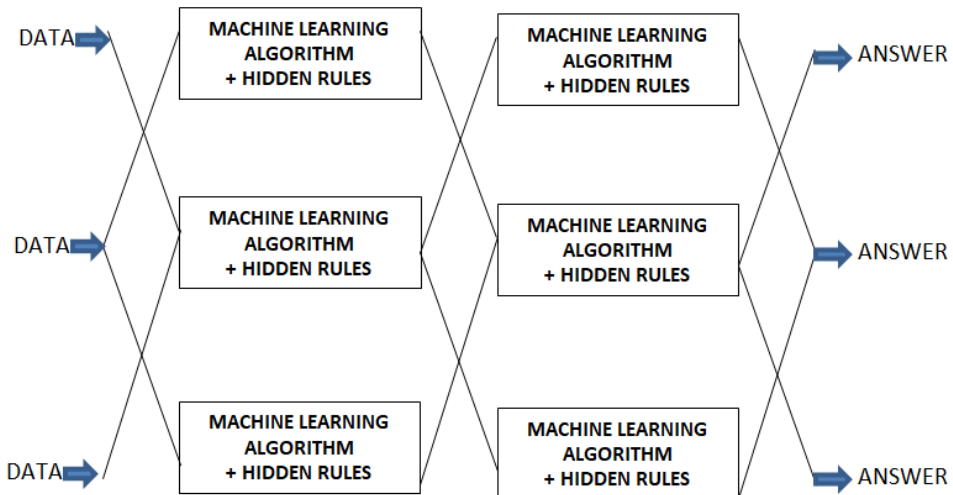
7. SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):

- i. This section has 16 questions.
- ii. A candidate has to do 10 questions.
- iii. Do as per the instructions given.
- iv. Marks allotted are mentioned against each question/part.

SECTION A: OBJECTIVE TYPE QUESTIONS

Q.1	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)	Mark
i.	Renuka wakes up every day at 5 AM, completes her homework and attends her dance classes, all by herself. No one tells her to do it, but she remains focused on her goal. Which self-management skill does Rekha demonstrate in this scenario?	1
ii.	Which of the following functions is not performed using a mouse? (a) Turn on (b) Hover (c) Right click (d) Drag and Drop	1
iii.	Assertion (A): Entrepreneurs are born with the qualities needed for business success. Reason (R): Entrepreneurial skills can be learned and developed over time through experience and education. (a) Both A and R are correct, and R is the correct explanation of A (b) Both A and R are correct, but R is NOT the correct explanation of A (c) A is correct, but R is not correct (d) A is not correct, but R is correct	1

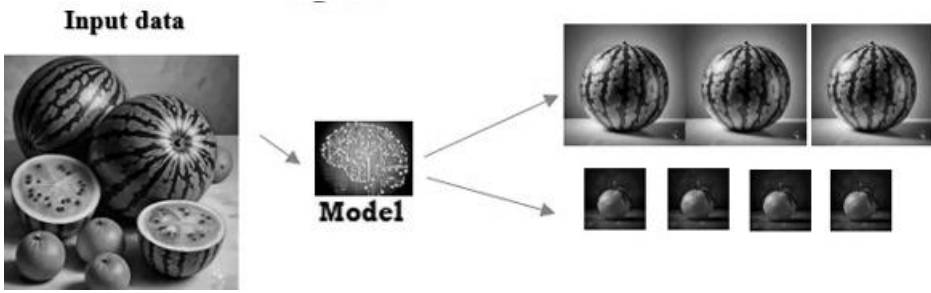
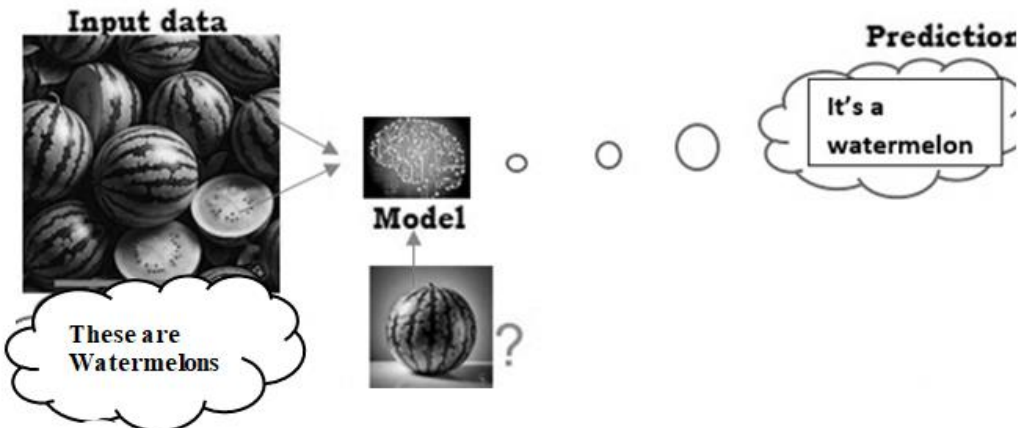
iv	_____ is the ability to control one's emotions, thoughts, and behaviors effectively in different situations. a) Mental Health b) Emotional Health c) Self-Management d) Stress Management	1
v.	Consider a situation where a manager is trying to convey important instructions to a team member through a text message. However, the team member misinterprets the tone and intention of the message. a. Interpersonal barrier b. Physical barrier c. Organisational barrier d. Linguistic barrier	1
vi.	Education plays a key role in sustainable development by helping individuals secure jobs, support their families, and become responsible citizens. Efforts such as using available educational facilities, helping friends with their studies, and preventing school dropouts are essential." Which SDG can you relate this statement to? a. Decent Work and Economic Growth b. No Poverty c. Quality Education d. Reduced Inequalities	1
Q.2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Assertion (A): Some students are good at mathematics, while others excel at music. Reason (R): Different individuals possess various forms of intelligence, each at varying levels. a. Both A and R are correct, and R is the correct explanation of A b. Both A and R are correct, but R is NOT the correct explanation of A c. A is correct, but R is incorrect d. A is incorrect, but R is correct	1
ii.	What term refers to the protection of personal information from misuse, especially regarding apps and online data? a. AI Ethics b. Data Privacy c. AI Bias d. AI Access	1
iii.	Consider the following statements about neural networks: Statement 1: Neural networks typically have multiple layers. Statement 2: The layer responsible for producing the final output of a neural network is the first layer. Which of the following is true? (a) Both Statement1 and Statement2 are correct (b) Both Statement1 and Statement2 are incorrect (c) Statement 1 is correct but Statement2 is incorrect (d) Statement2 is correct but Statement1 is incorrect.	1

iv	Observe the representation. and fill in the blank:  The job of an input layer is _____	1
v.	Expansion of DBMS is _____ (a) Data Management System (b) Database Manages Systematic (c) Data board Managing System (d) Database Management System	1
vi.	If the words have highest occurrence in all the documents of the corpus, they are said to have negligible value hence they are termed as _____. (a) Stop words (b) Rare word (c) Frequent word (d) Common word	1
Q.3	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Consider the following examples: Example 1: Using a language model to generate text Example 2: Providing recommendations to the users of platforms like Netflix, Amazon, Spotify, YouTube etc. Example 3: Monitoring the physical and mental health using chatbots and other health apps. Example 4: Dispensing hand sanitizer automatically Which of these examples do not involve artificial intelligence? a. Example 1 and Example 2 b. Example 3 c. Example 4 d. Example 1, Example 3, and Example 4	1
ii.	_____ is known as the stage of acquiring data for the project.	1

iii.	Which of the following is NOT an application of data science? a. Fraud and Risk Detection in Finance b. Genetics & Genomics c. Targeted Advertising d. Face lock in smart phone	1
iv	_____ is a computer vision task of assigning/categorizing an input image one label from a fixed set of categories. a. Instance segmentation b. Object detection c. Image Classification d. Image segmentation	1
v.	Identify the given Chatbot type: It follows a predefined script and is limited in its ability to understand and respond to complex queries of the humans.	1
vi.	What is the formula for calculating the F1 score? a. $F1 = (\text{precision} + \text{recall}) / 2$ b. $F1 = \text{precision} * \text{recall} / 2$ c. $F1 = 2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$ d. $F1 = \text{precision} / \text{recall}$	1
Q.4	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Which type of machine learning involves using a labeled dataset to train the model? (a) Unsupervised Learning (b) Supervised Learning (c) Dimensionality Reduction (d) Deep Learning	1
ii.	Assertion (A): The confusion matrix is an evaluation metric used to measure the performance of a prediction model. Reason (R): Accuracy is calculated by dividing the sum of True Positives (TP) and True Negatives (TN) by the total number of observations. a. Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A). b. Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). c. Assertion (A) is true, but Reason (R) is false. d. Assertion (A) is false, but Reason (R) is true.	1
iii.	A movie theater wants to analyze its ticket sales and refreshment purchases, both offline and online, to identify peak sales times and determine the most popular refreshments. The theater aims to use this data to make predictions for better stocking and staffing decisions. Which of the following is the most appropriate method for collecting relevant data for analysis? a. Conducting surveys at the ticket and refreshment counters to gather customer preferences. b. Searching government portals to obtain ticket sales data. c. Visiting private websites to gather data from other theaters. d. Visiting a nearby theater to compare sales records	1

iv	All the colors that are present can be made by combining different intensities of three primary colors. What are they?	1
v.	What does the Bag of Words model provide when applied to a text corpus? a. The sequence of words in the corpus and their synonyms b. A vocabulary of unique words and their frequency of occurrence c. The grammatical structure of the sentences in the corpus d. A list of sentences and their lengths in the corpus	1
vi.	In a confusion matrix, which value represents the correct positive predictions made by a model? a. True Positive b. False Positive c. True Negative d. False Negative	1
Q.5	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	You're tasked with developing an AI system that can automatically analyze medical images to detect anomalies. Which field of artificial intelligence would be most relevant for this task? a. Natural Language Processing b. Computer Vision c. Machine Learning d. Data Science	1
ii.	In a confusion matrix, which value represents the correct negative predictions made by a model?	1
iii.	In pixel value, _____ value is to be taken as no colour or black. (a) 255 b) 1 c) 0 d) 3	1
iv	Which of the following describes tokenization in Natural Language Processing (NLP)? a. Splitting text into smaller units like words or phrases b. Processing images into pixels c. Creating labelled datasets d. Detecting patterns in large data	1
v.	In the process of text normalization, which technique aims to reduce words to their root form while ensuring that the resulting word remains meaningful? a. Tokenization b. Stemming c. Segmentation d. Lemmatization	1
vi.	Which of the following best defines Accuracy in a classification model? a. The percentage of correct predictions out of all observations made by the model. b. The proportion of positive cases that were correctly predicted. c. The percentage of correct negative predictions out of all predictions. d. The percentage of False Positives and True Negatives.	1

	SECTION B: SUBJECTIVE TYPE QUESTIONS																																											
	Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks) Answer each question in 20 – 30 words.																																											
Q.6	List any four barriers that affect effective communication.										2																																	
Q.7	What is Time Management?										2																																	
Q.8	Anuska is searching for sports-related content on a search engine. How can she protect her data while using the internet?										2																																	
Q.9	List two qualities of successful Entrepreneurs.										2																																	
Q.10	What are the major problems of related to sustainable development?										2																																	
	Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)																																											
Q.11	How do you understand whether a machine/application is AI based or not? Explain with an example.										2																																	
Q.12	Sam wants to create a grading system where students are classified based on the grades they receive according to their exam marks. Which learning model should he use, and why?										2																																	
Q.13	Ajmal, a data scientist collected data in the form of tables and it is stored in different formats. Write any two formats, that he can store the tabular data.										2																																	
Q.14	Describe the term pixel value and what are its possible values.										2																																	
Q.15	Explain any one application of Natural Language Processing which are used in the real-life scenario										2																																	
Q.16	According to the data given below, Calculate TP,TN,FP,&FN.										2																																	
	<table><tr><td>Index</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>Actual</td><td>Cat</td><td>Cat</td><td>Cat</td><td>Not Cat</td><td>Not Cat</td><td>Cat</td><td>Not Cat</td><td>Cat</td><td>Not Cat</td><td>Cat</td></tr><tr><td>Predicted</td><td>Cat</td><td>Cat</td><td>Not Cat</td><td>Cat</td><td>Not Cat</td><td>Cat</td><td>Cat</td><td>Not Cat</td><td>Cat</td><td>Not Cat</td></tr></table>											Index	1	2	3	4	5	6	7	8	9	10	Actual	Cat	Cat	Cat	Not Cat	Not Cat	Cat	Not Cat	Cat	Not Cat	Cat	Predicted	Cat	Cat	Not Cat	Cat	Not Cat	Cat	Cat	Not Cat	Cat	Not Cat
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	Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)																																											
Q.17	Sangeetha attended a seminar on Artificial Intelligence and has now been asked to write a report on his learnings from the seminar. Being a non-technical person, he understood that the AI enabled machine uses data of different formats in many of the daily based applications but failed to sync it with the right terminologies and express the details. Help Sangeetha to define Artificial intelligence and list the three domains of AI and the data that is used in these domains.										4																																	
Q.18	Jack wants to learn how to acquire data for an AI Project. a) Explain him the various ways to collect the data. b) Differentiate between test data and training data										4																																	

Q.19	Identify and explain the types of the learning-based approaches in the figures given below. Figure 1  Figure 2 	4													
Q.20	Explain how humans process language and compare it with how computers process language. What challenges do computers face in understanding human language?	4													
Q.21	The country was shaken up by a series of earthquake which has done a huge damage to the people as well as the infrastructure. To address this issue, an AI model has been created which can predict if there is a chance of a earthquake or not. The confusion matrix for the same is as below: <table data-bbox="402 1431 1077 1608"> <tr> <th colspan="2" rowspan="2">Confusion matrix</th><th colspan="2">Reality</th></tr> <tr> <th>YES</th><th>NO</th></tr> <tr> <th rowspan="2">prediction</th><th>YES</th><td>150</td><td>40</td></tr> <tr> <th>NO</th><td>50</td><td>60</td></tr> </table> a) How many total tests have been performed in the above scenario. b) Calculate precision, recall and F1 score.	Confusion matrix		Reality		YES	NO	prediction	YES	150	40	NO	50	60	4
Confusion matrix				Reality											
		YES	NO												
prediction	YES	150	40												
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End of paper